

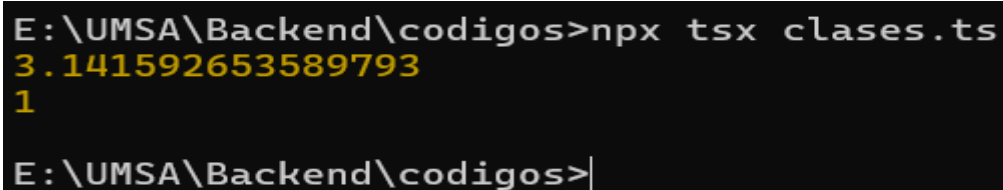
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MATERIA: DESARROLLO WEB BACKEND

1. Agregar los getters de radio

```
class Circulo {  
    public constructor(public readonly radio: number) {}  
  
    public area(): number {  
        return Math.PI * this.radio * this.radio;  
    }  
  
    // Getter de radio  
    public getRadio(): number {  
        return this.radio;  
    }  
}  
  
const circulo = new Circulo(1);  
console.log(circulo.area());  
console.log(circulo.getRadio());
```

CORRIDA:



```
E:\UMSA\Backend\codigos>npx tsx clases.ts  
3.141592653589793  
1  
E:\UMSA\Backend\codigos>|
```

2. Agregar la clase padre Figura (color: string) forma corta

```
class Figura {
```

```

        constructor(public color: string) {}
    }

class Circulo extends Figura {
    public constructor(public readonly radio: number, color: string) {
        super(color);
    }

    public area(): number {
        return Math.PI * this.radio * this.radio;
    }

    public getRadio(): number {
        return this.radio;
    }
}

const circulo = new Circulo(1, 'rojo');
console.log(circulo.area());
console.log(circulo.getRadio());
console.log(circulo.color);

```

CORRIDA:

```

E:\UMSA\Backend\codigos>npx tsx clases.ts
3.141592653589793
1
rojo
E:\UMSA\Backend\codigos>|

```

3. Ámbito de acción de los atributos (públicos, protegidos y privados)

```

class Figura {
    public color: string;           protected tipo: string;
    private id: number;
}

```

```
    constructor(color: string) {
        this.color = color;
        this.tipo = "Figura";
        this.id = Math.random();
    }

    public getId(): number {
        return this.id;
    }
}

class Circulo extends Figura {
    public readonly radio: number;

    constructor(radio: number, color: string) {
        super(color);
        this.radio = radio;
        this.tipo = "Círculo";
    }

    public getRadio(): number {
        return this.radio;
    }

    public getTipo(): string {
        return this.tipo;
    }
}

const circulo = new Circulo(5, "Rojo");
console.log(circulo.color);
console.log(circulo.getRadio());
console.log(circulo.getTipo());
```

CORRIDA:

```
E:\UMSA\Backend\codigos>npx tsx clases.ts
0.05545243867462191
Círculo
1
E:\UMSA\Backend\codigos>
```